THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE UNIT TEST SET A

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. Which cell does NOT have a nucleus when mature? 1

(a) onion peel cell **(b)** human nerve cell
(c) cheek cell **(d)** human red blood cell

2. The control centre of a eukaryotic cell is the: 1

(a) vacuole **(b)** Golgi apparatus **(c)** nucleus **(d)** ribosome

3. The ability of a plant cell to grow into a complete plant is called: 1

(a) plasmolysis **(b)** mitosis **(c)** osmosis **(d)** totipotency

4. Bones are connected to muscles by: 1

(a) nerves **(b)** cartilage **(c)** tendons **(d)** ligaments

5. The basic unit of nervous tissue is the: 1

(a) nucleus **(b)** neuron **(c)** fibre **(d)** nephron

6. Coconut husk fibres used for mats are made of: 1

(a) collenchyma **(b)** parenchyma **(c)** phloem **(d)** sclerenchyma

7. Amoeba, Paramecium and Euglena belong to the kingdom: 1

(a) Monera **(b)** Fungi **(c)** Animalia **(d)** Protista

8. Animals with jointed legs and an exoskeleton belong to: 1

(a) Mollusca **(b)** Annelida **(c)** Arthropoda **(d)** Echinodermata

9. Pollination between two flowers of different plants is: 1

(a) self-pollination **(b)** fertilisation
(c) budding **(d)** cross-pollination

10. Why do we refrigerate food? 1

(a) Low temperature slows down microbial growth
(b) it makes spores
(c) it kills all bacteria
(d) it adds nutrients

11. A wavelength of 100 nm in metres is: 1

(a) 100×10^{-3} m **(b)** 100×10^{-9} m **(c)** 100×10^{-6} m **(d)** 100×10^{-12} m

12. Electromagnetic waves: 1

(a) are longitudinal only **(b)** are slower than sound

(c) need a medium **(d)** can travel through vacuum

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 10–15 words each.

13. Differentiate between unicellular and multicellular organisms with examples. 2

14. Why do meristematic cells lack large vacuoles? 2

15. Name any four characteristics used for classifying organisms. 2

16. State Pasteur's contribution to the study of reproduction. 2

17. How much solar energy falls on 1 m^2 in one hour at an insolation of 1 kW m^{-2}? 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 30–40 words each.

18. Describe an activity to demonstrate osmosis using potato pieces. 3

19. State three advantages of classification. 3

20. Why does sexual reproduction lead to variation? 3

21. Explain how a change in the cryosphere can affect the other spheres. 3

22. Name three artificial methods of vegetative propagation and one benefit. 3

OR

Describe the development of a fruit and seed after fertilisation. 3

23. Describe the layers of the atmosphere with one feature each. 3

OR

Why is India suitable for solar energy? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

24. (i) Compare xylem and phloem in a table of four points. 5
(ii) How does water rise to the top of tall trees?

OR

(i) Describe the three types of meristematic tissues. 5
(ii) Explain with an experiment that roots grow at their tips.
(iii) State the features of meristematic cells.

25. (i) Why is the heating of the Earth uneven? 5
(ii) How does it drive winds?
(iii) Explain land and sea breezes.

OR

(i) Explain the greenhouse effect. 5
(ii) What is global warming?
(iii) State four consequences.

END OF QUESTION PAPER Total: 50 Marks